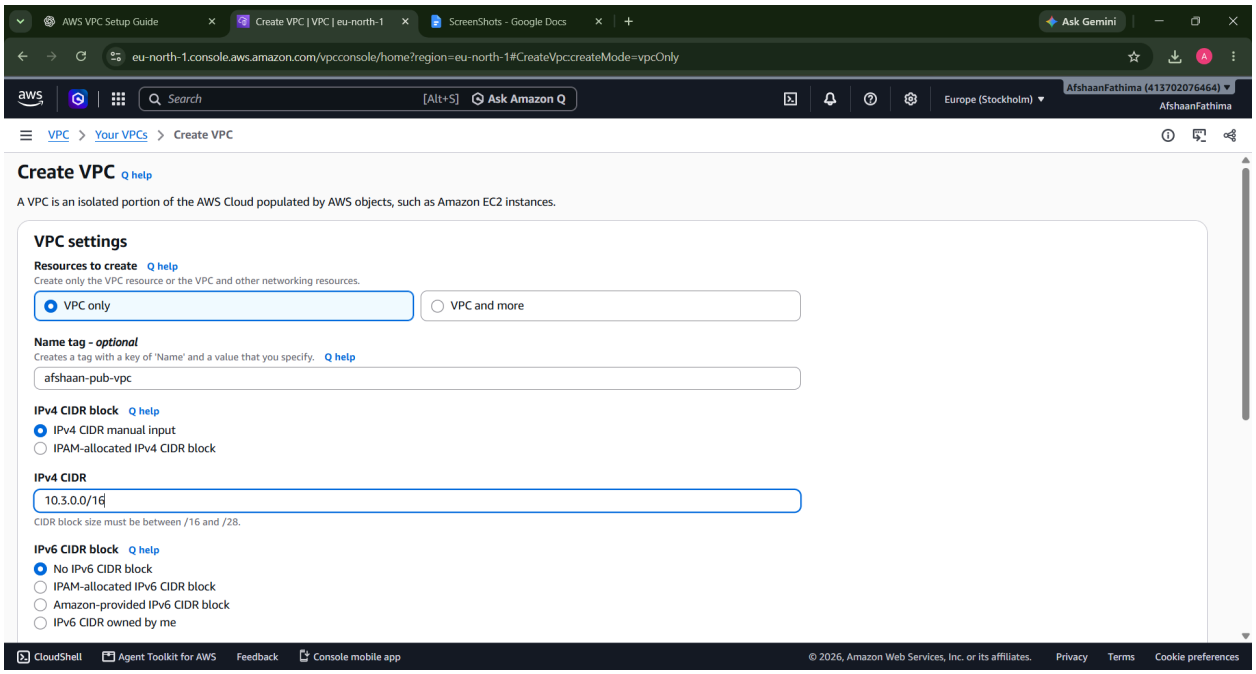
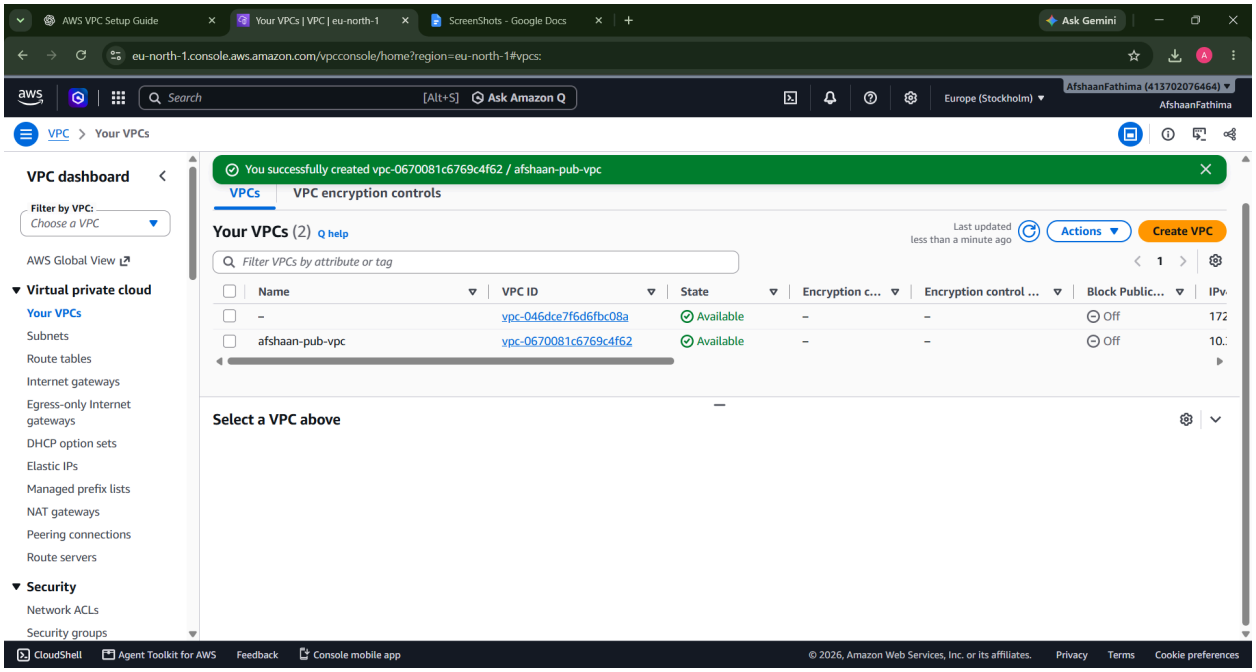
CONFIGURATION OF VPC

Open your browser.Go to Aws console
Sign in and In the top search bar, type VPC

Open Your VPCs

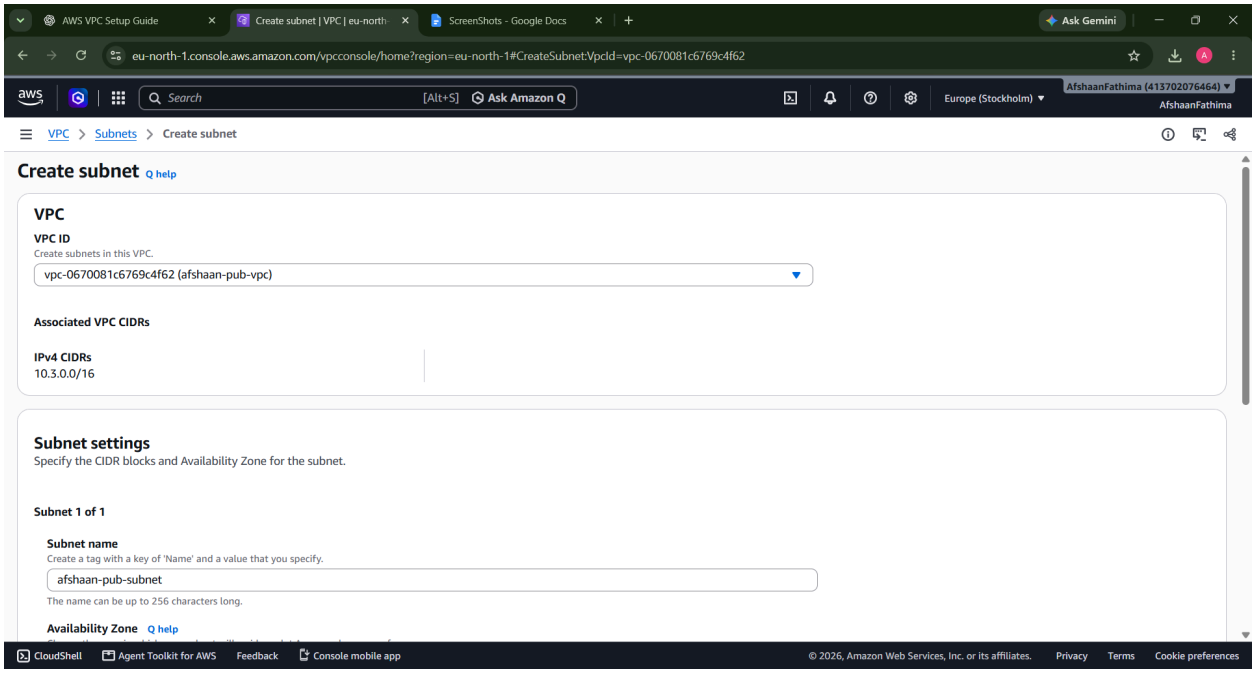
1. On the left navigation panel, select Your VPCs.
2. Click the orange Create VPC button.

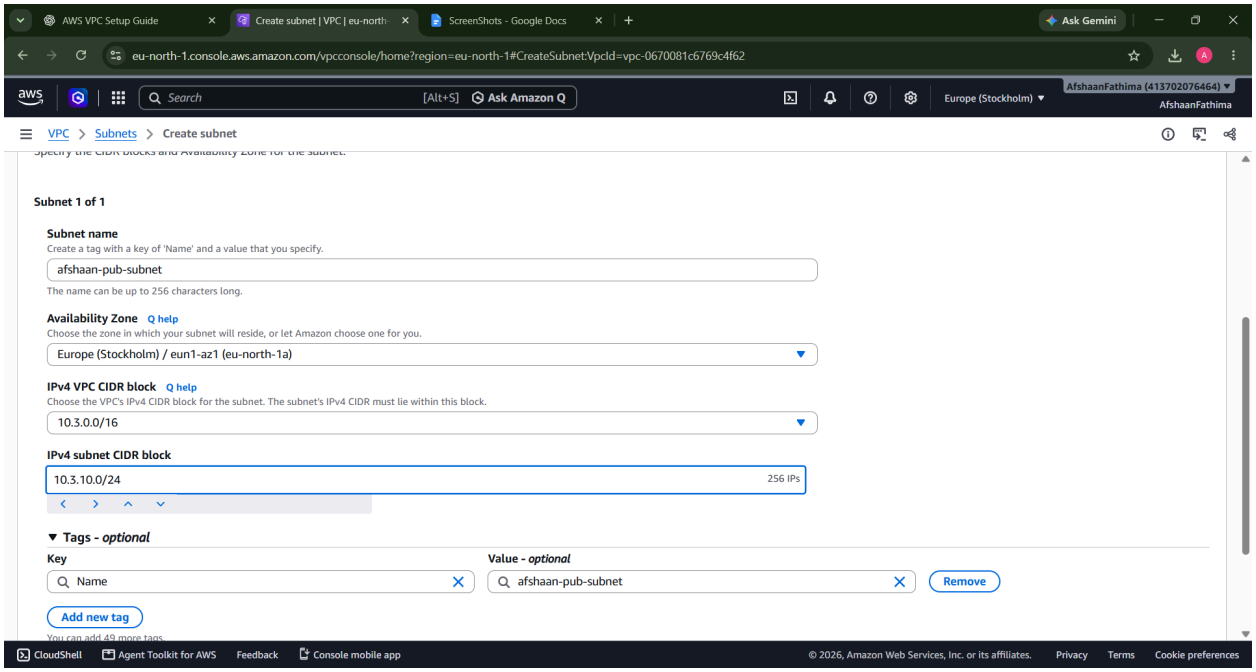




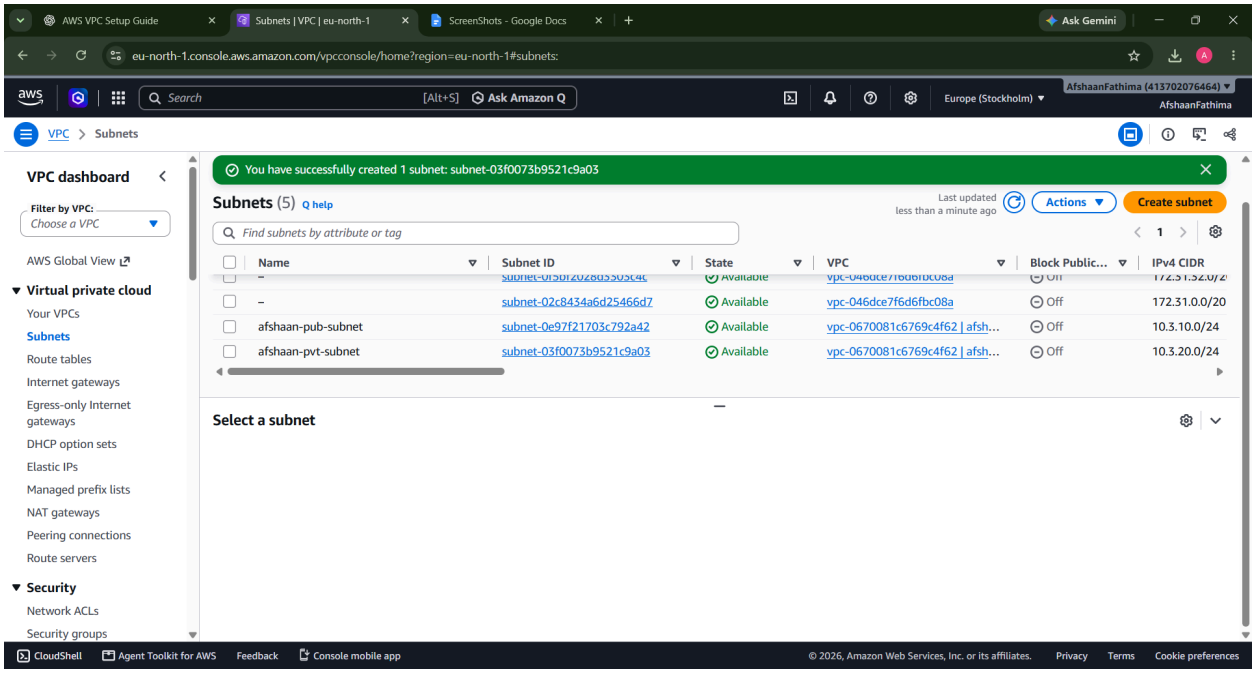
Step 2 - Create the Public Subnet

Go to the AWS VPC Console.On the left navigation panel, find Subnets under Virtual private cloud.Click Subnets.Click Create subnet.



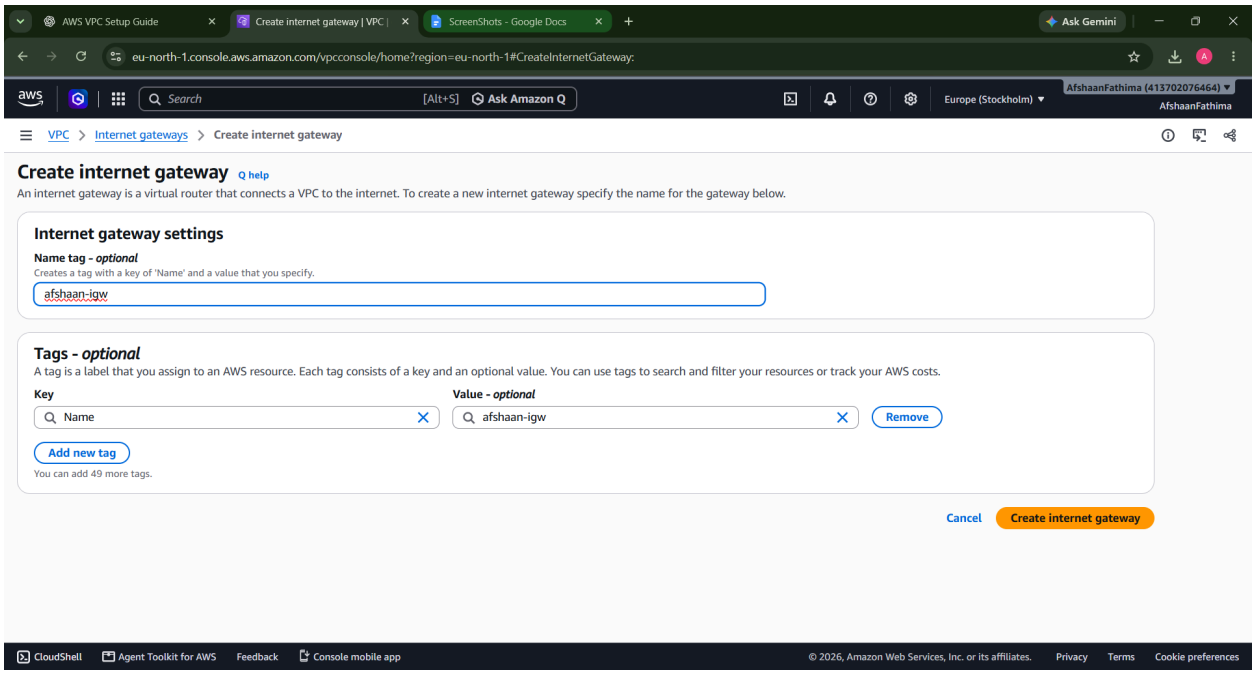


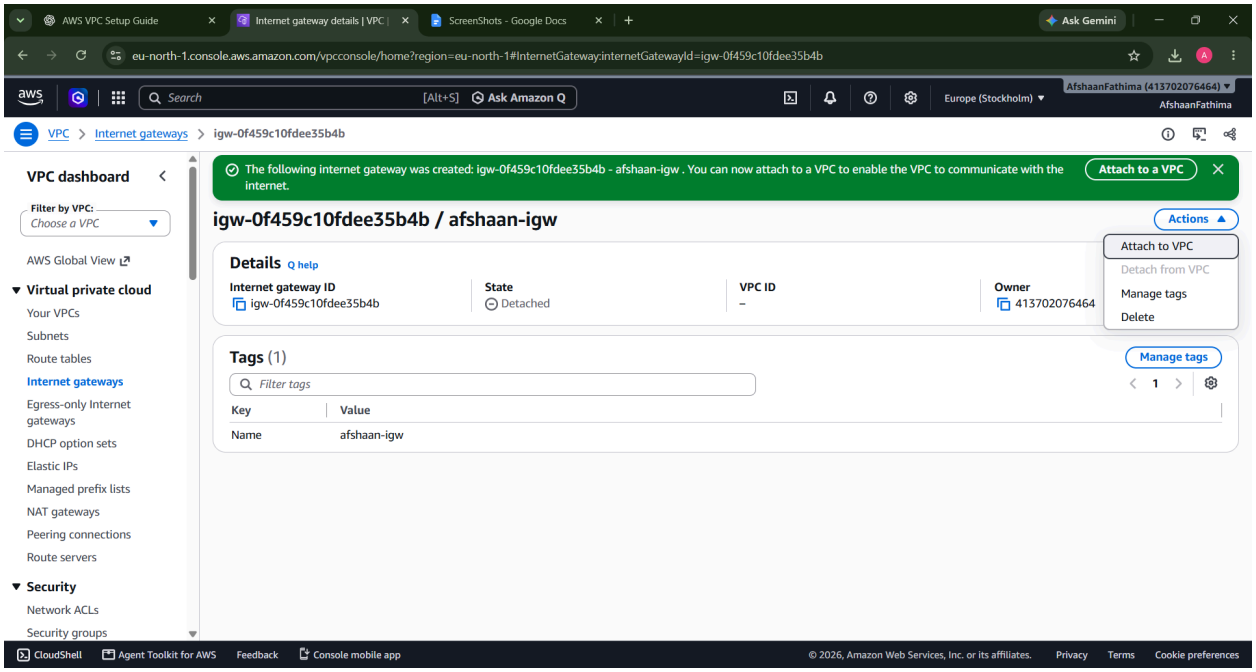
Step 3 - Create the Private Subnet
Name: afshaan-pvt-subnet
VPC: afshaan-pub-vpc
CIDR: 10.3.20.0/24

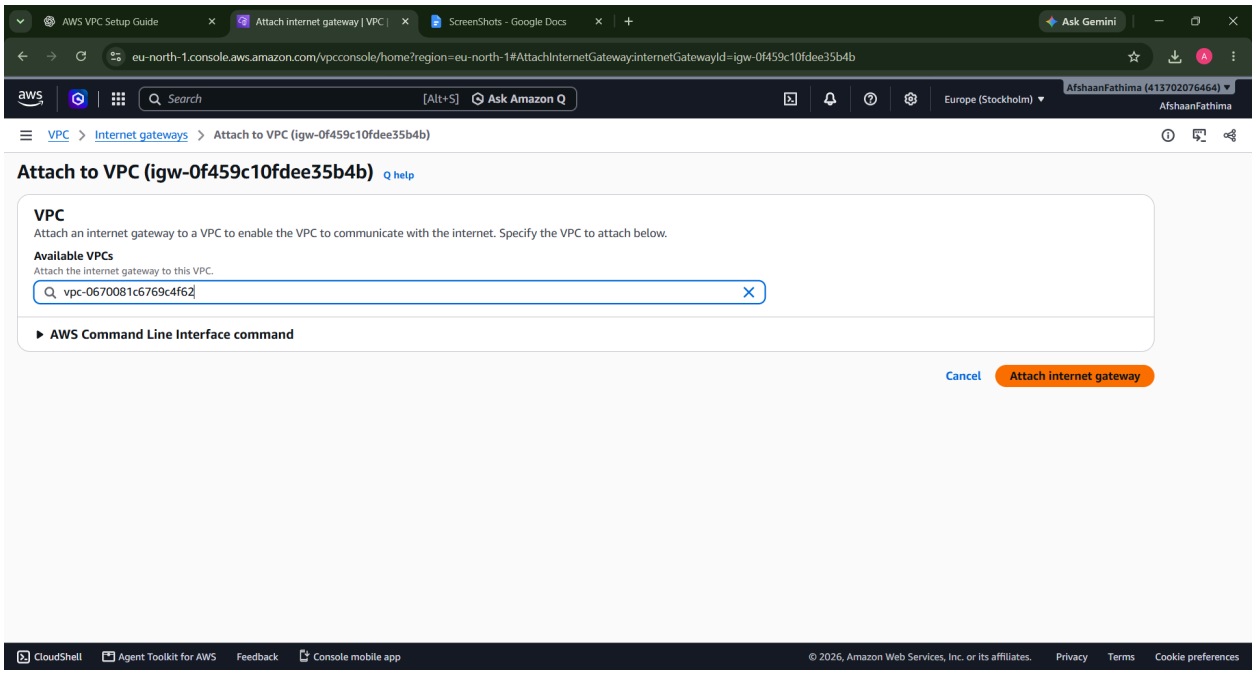


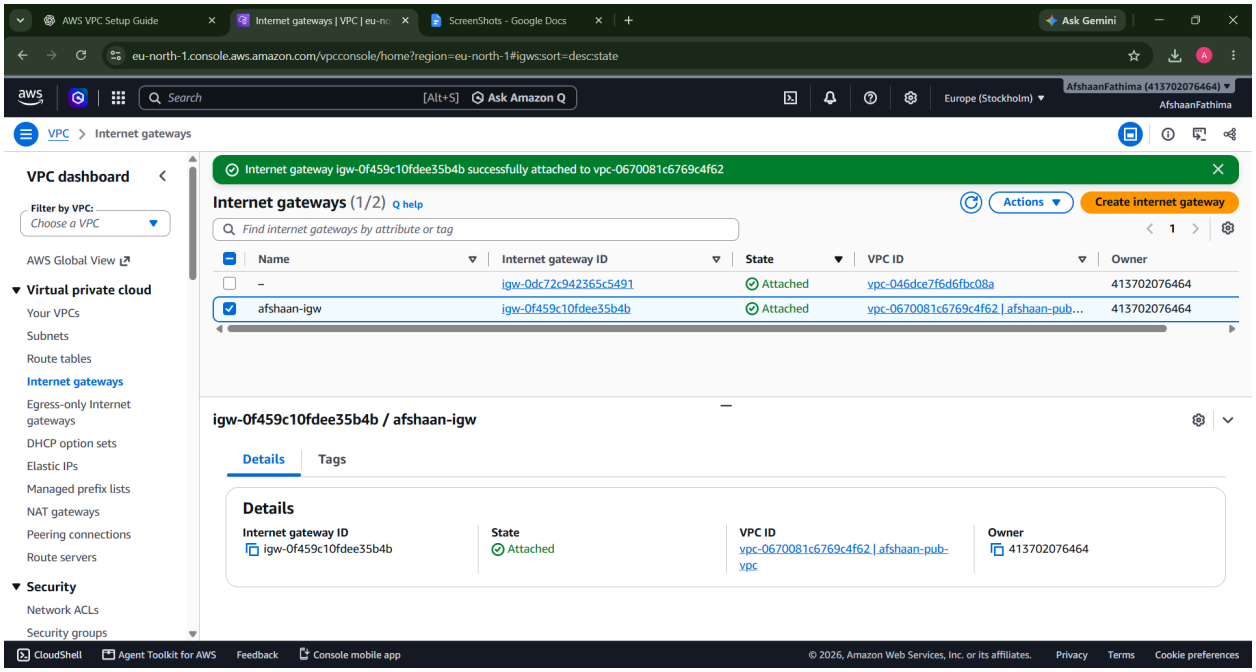
Step 4 - Create and Attach Internet Gateway

On the left navigation panel, find Internet gateways.Click Internet gateways.Click Create internet gateway.

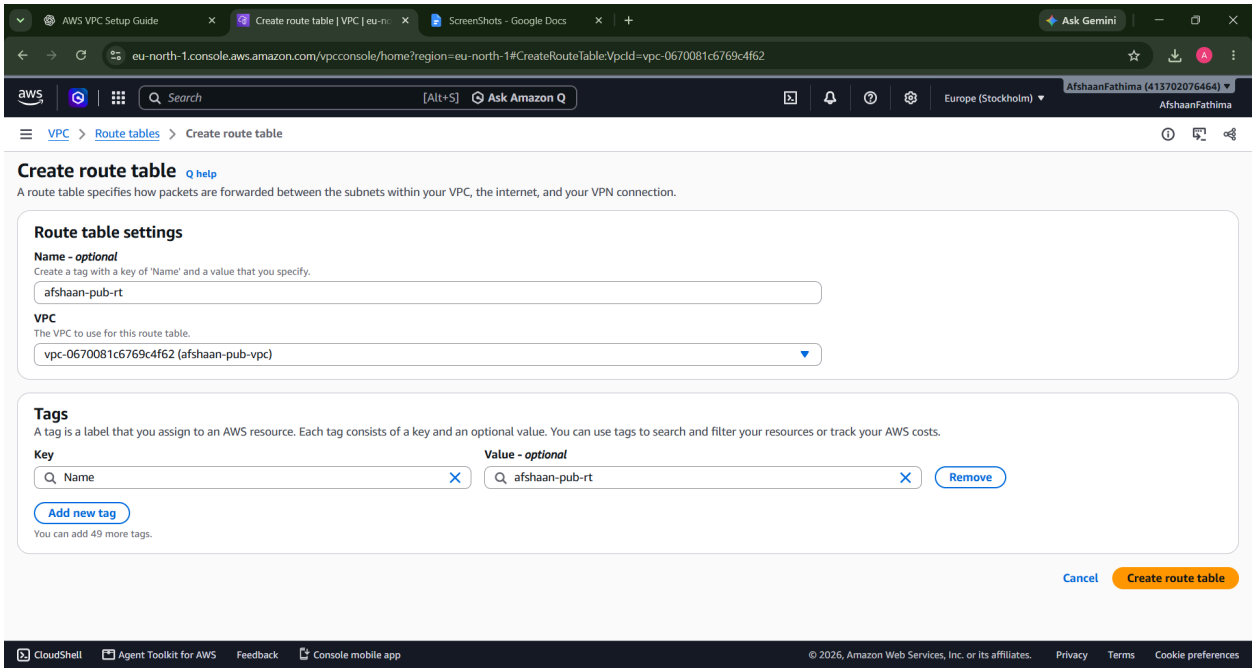


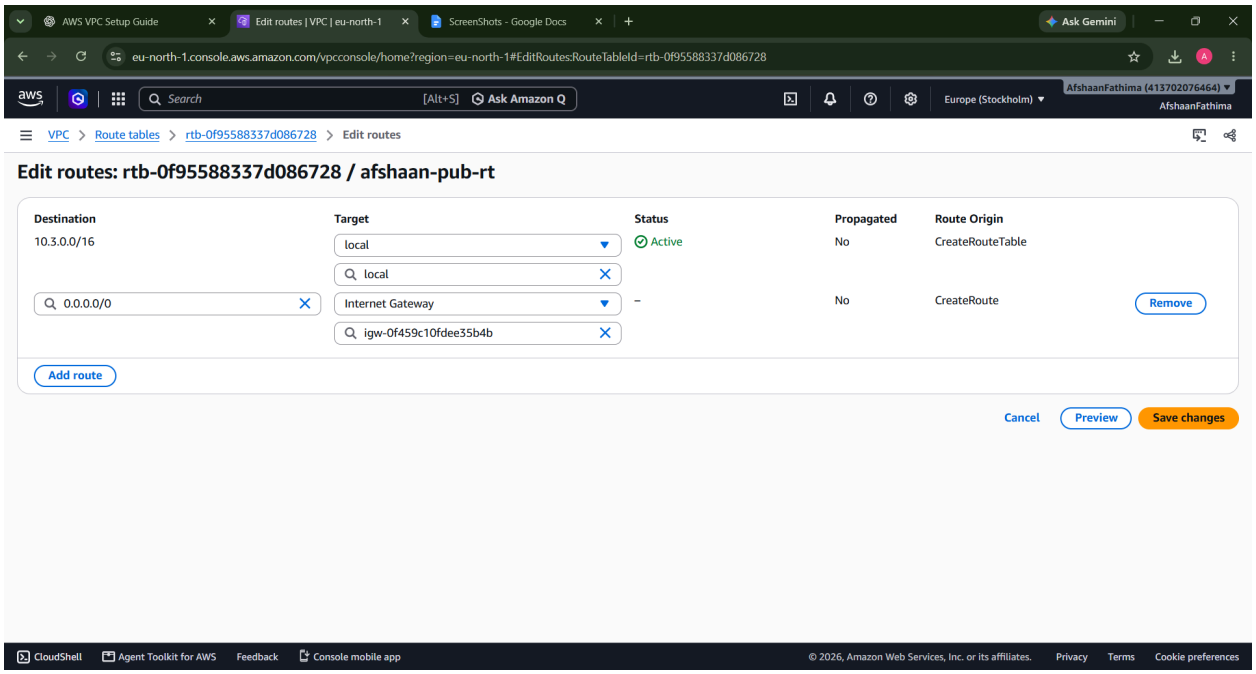

AWS VPC Setup Guide
Internet gateway details | VPC
ScreenShots - Google Docs
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#InternetGateway:internetGatewayId=igw-0f459c10fdee35b4b
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
AfshaanFathima (413702076464)
AfshaanFathima
VPC > Internet gateways > igw-0f459c10fdee35b4b
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
The following internet gateway was created: igw-0f459c10fdee35b4b - afshaan-igw . You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC
igw-0f459c10fdee35b4b / afshaan-igw
Actions
Attach to VPC
Detach from VPC
Manage tags
Delete
Details
Internet gateway ID
igw-0f459c10fdee35b4b
State
Detached
VPC ID
–
Owner
413702076464
Tags (1)
Manage tags
Filter tags
Key
Value
Name
afshaan-igw
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

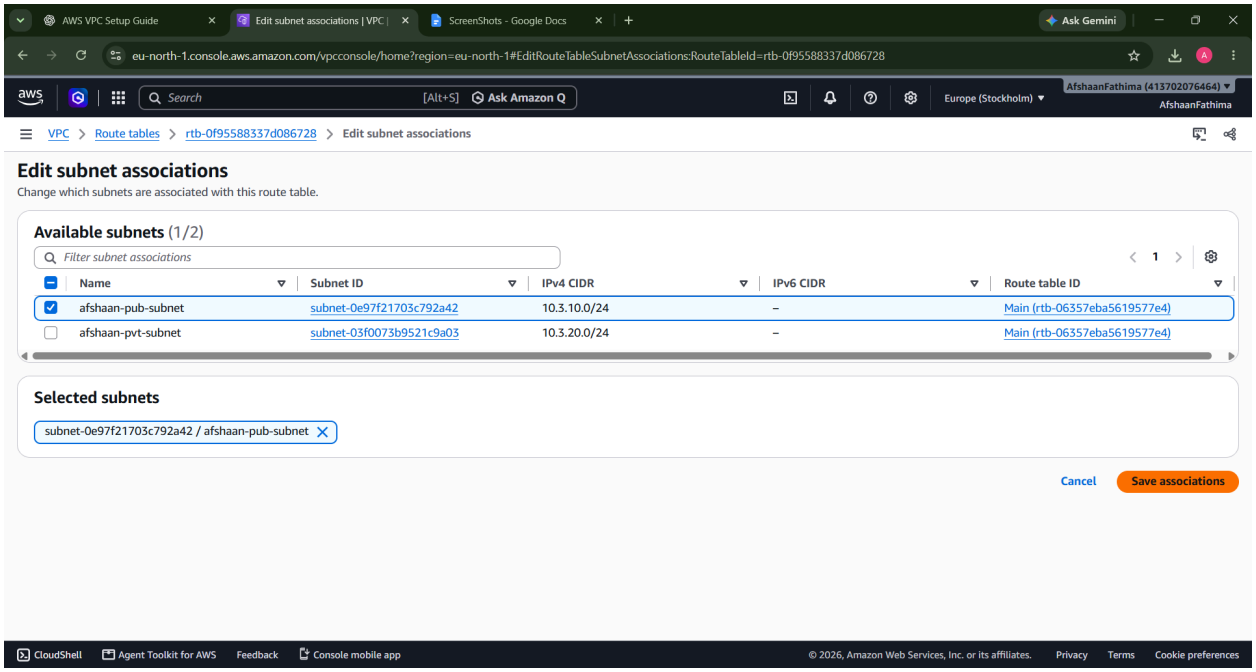

AWS VPC Setup Guide
Attach internet gateway | VPC
ScreenShots - Google Docs
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#AttachInternetGateway:internetGatewayId=igw-0f459c10fdee35b4b
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
AfshaanFathima (413702076464)
AfshaanFathima
VPC > Internet gateways > Attach to VPC (igw-0f459c10fdee35b4b)
Attach to VPC (igw-0f459c10fdee35b4b)
Q help
VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the internet gateway to this VPC.
vpc-0670081c6769c4f62
AWS Command Line Interface command
Cancel
Attach internet gateway
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

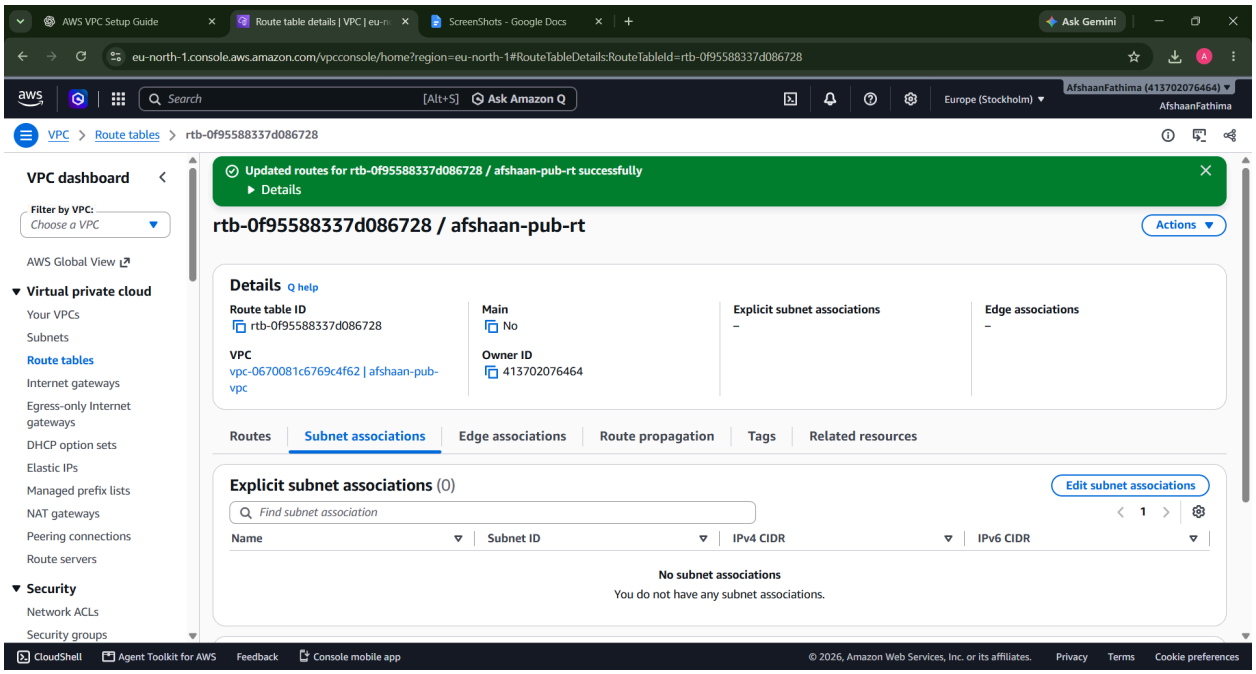


Step 5 - Create and Configure Public Route Table

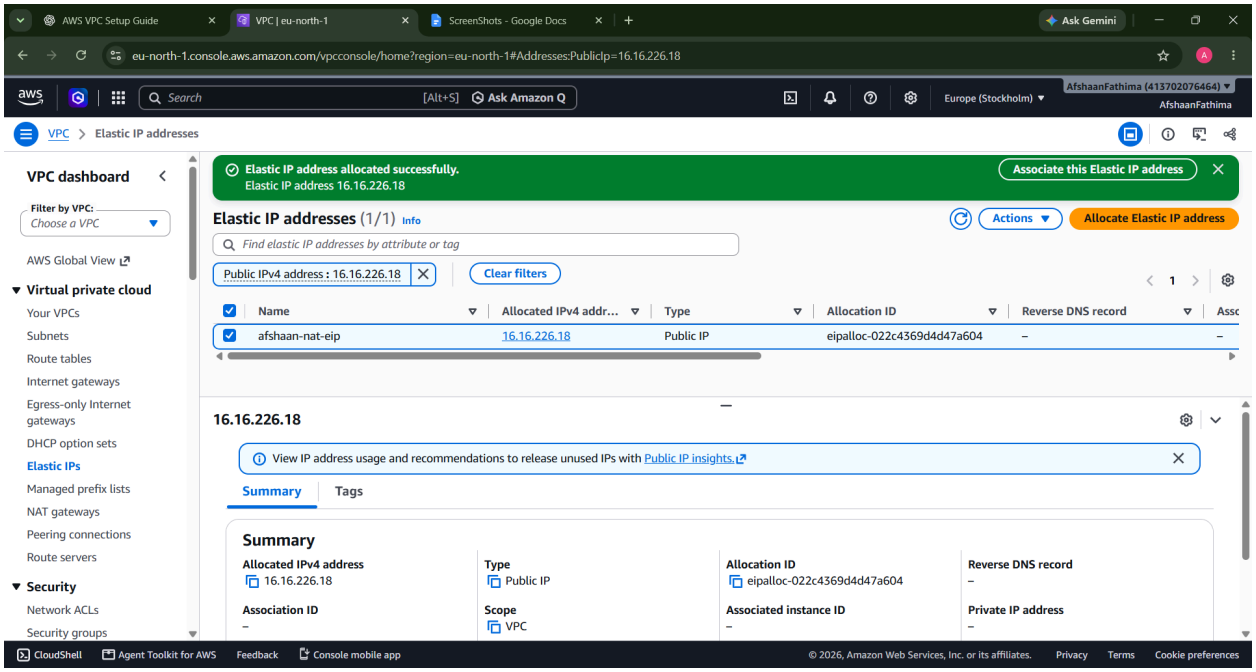

AWS VPC Setup Guide
Create route table | VPC | eu-n
ScreenShots - Google Docs
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#CreateRouteTable:VpcId=vpc-0670081c6769c4f62
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
AfshaanFathima (413702076464)
AfshaanFathima
VPC > Route tables > Create route table
Create route table
Q help
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
afshaan-pub-rt
VPC
The VPC to use for this route table.
vpc-0670081c6769c4f62 (afshaan-pub-vpc)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
afshaan-pub-rt
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


AWS VPC Setup Guide
Edit routes | VPC | eu-north-1
ScreenShots - Google Docs
Ask Gemini
eu-north-1.console.aws.amazon.com/vpcconsole/home?region=eu-north-1#EditRoutes:RouteTableId=rtb-0f95588337d086728
Search
[Alt+S]
Ask Amazon Q
Europe (Stockholm)
AfshaanFathima (413702076464)
AfshaanFathima
VPC > Route tables > rtb-0f95588337d086728 > Edit routes
Edit routes: rtb-0f95588337d086728 / afshaan-pub-rt
Destination
Target
Status
Propagated
Route Origin
10.3.0.0/16
local
Active
No
CreateRouteTable
local
0.0.0.0/0
Internet Gateway
-
No
CreateRoute
Remove
igw-0f459c10fdee35b4b
Add route
Cancel
Preview
Save changes
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

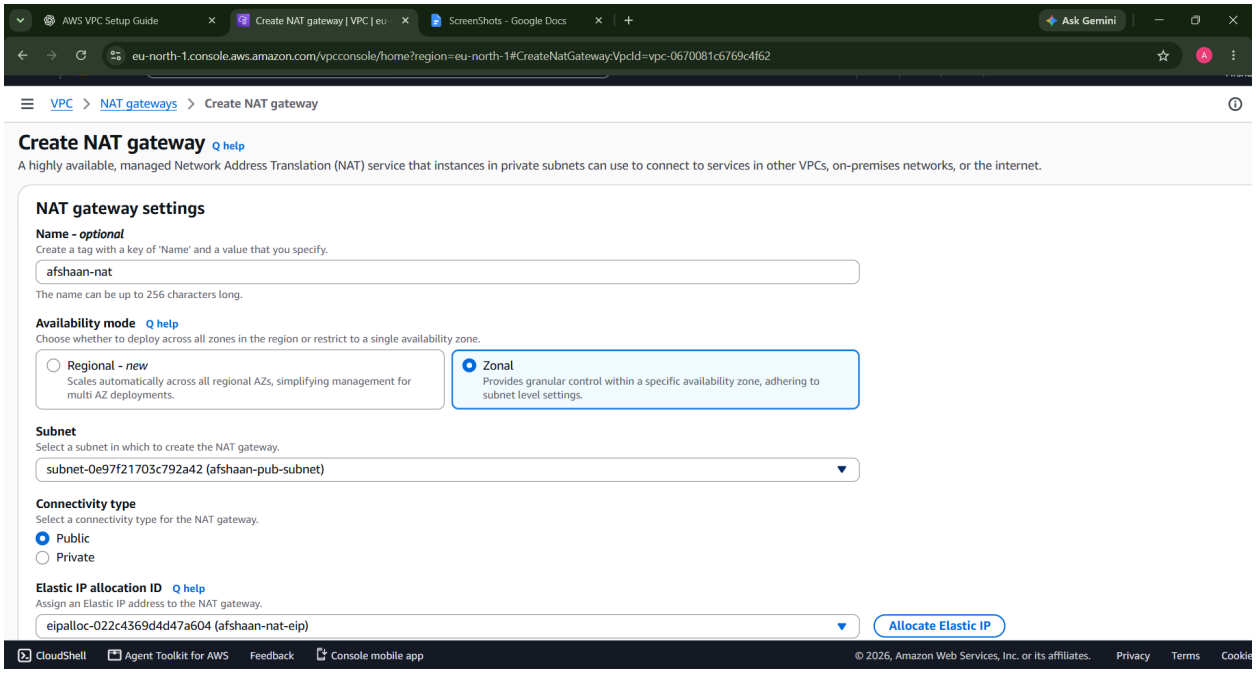


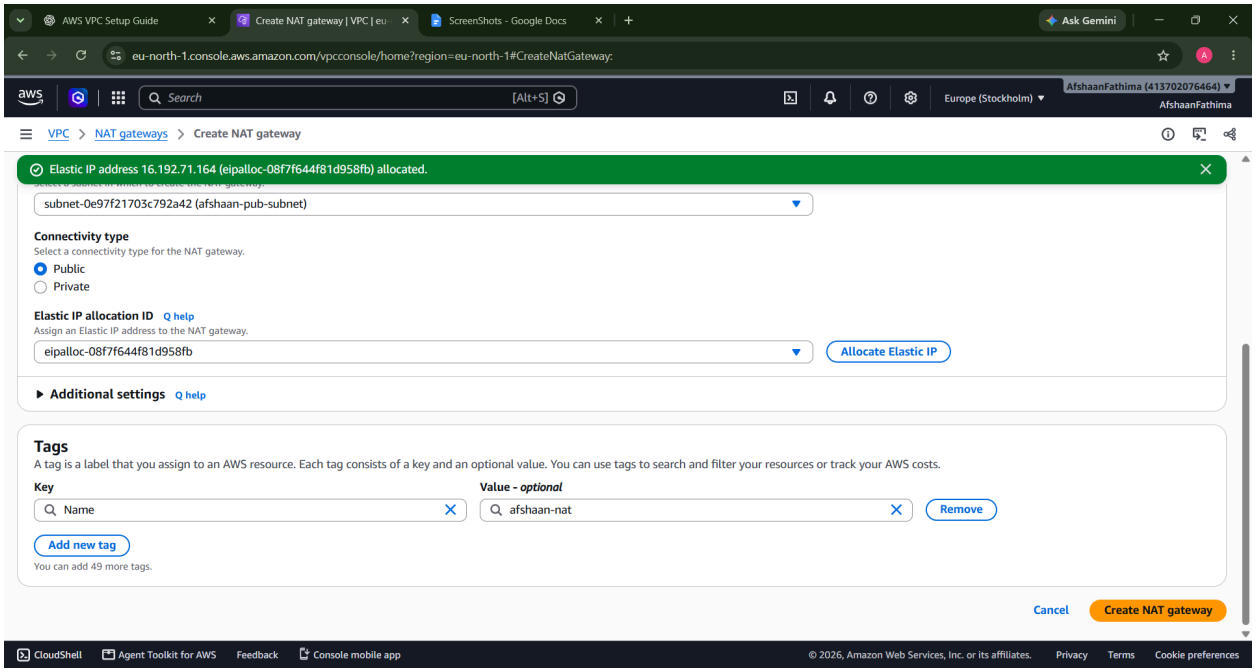


Allocate an Elastic IP

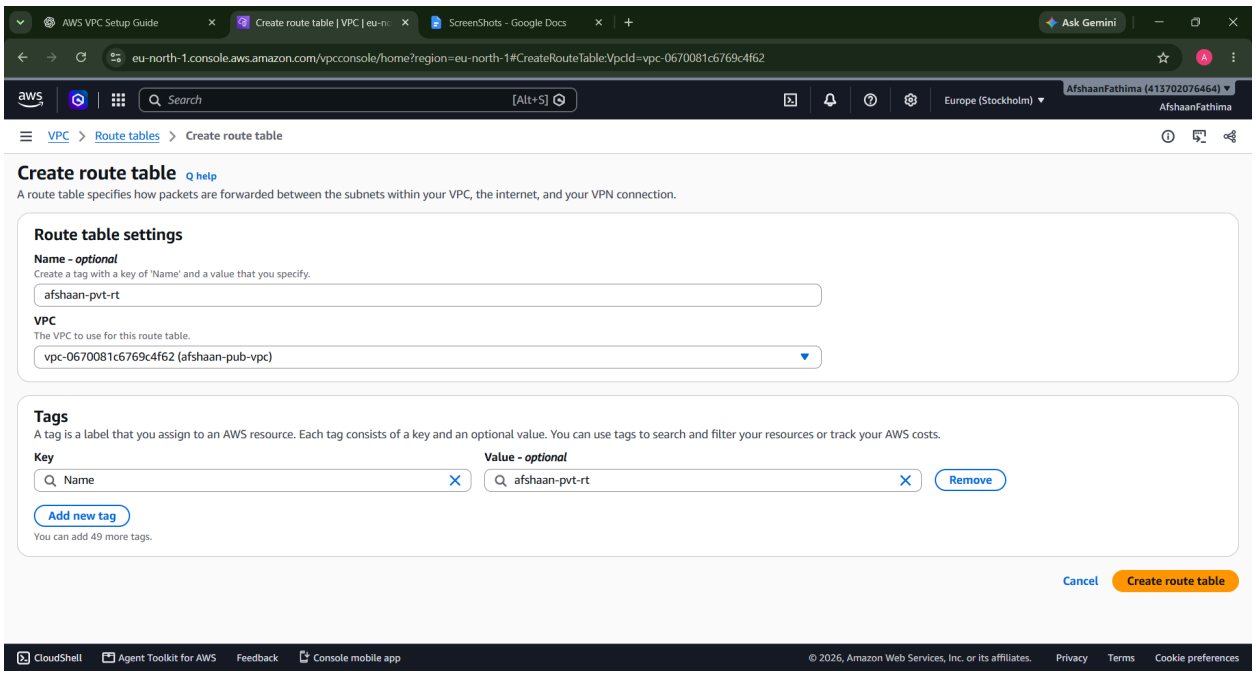


Create NAT Gateway

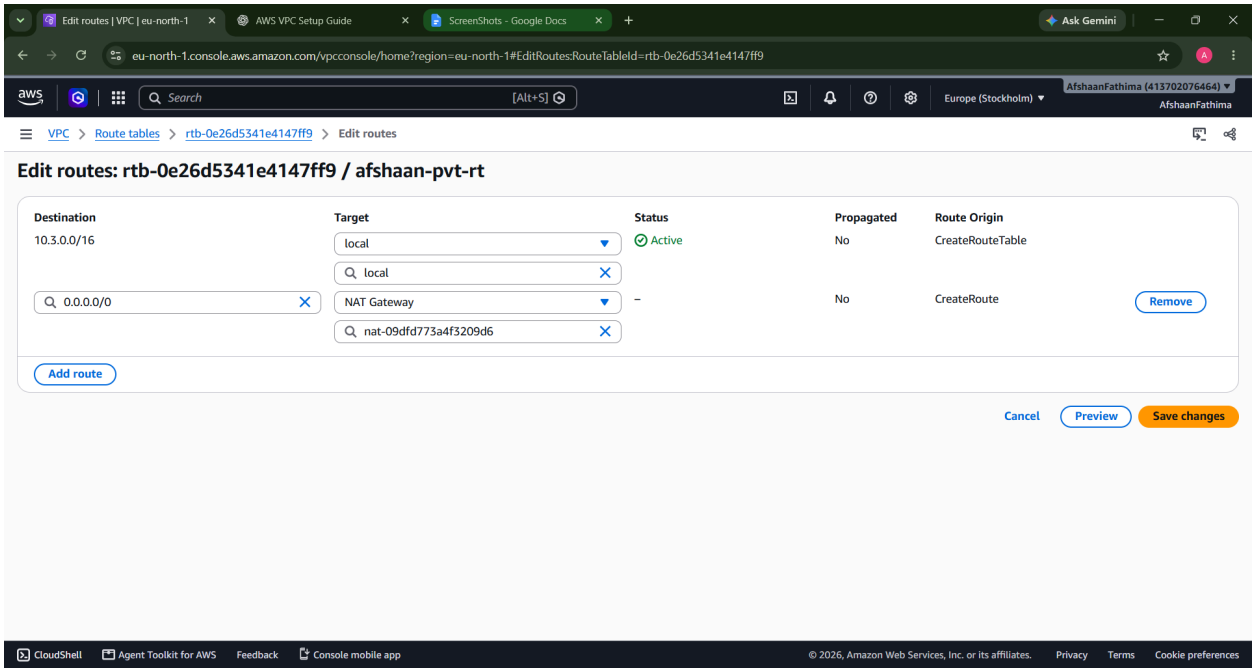




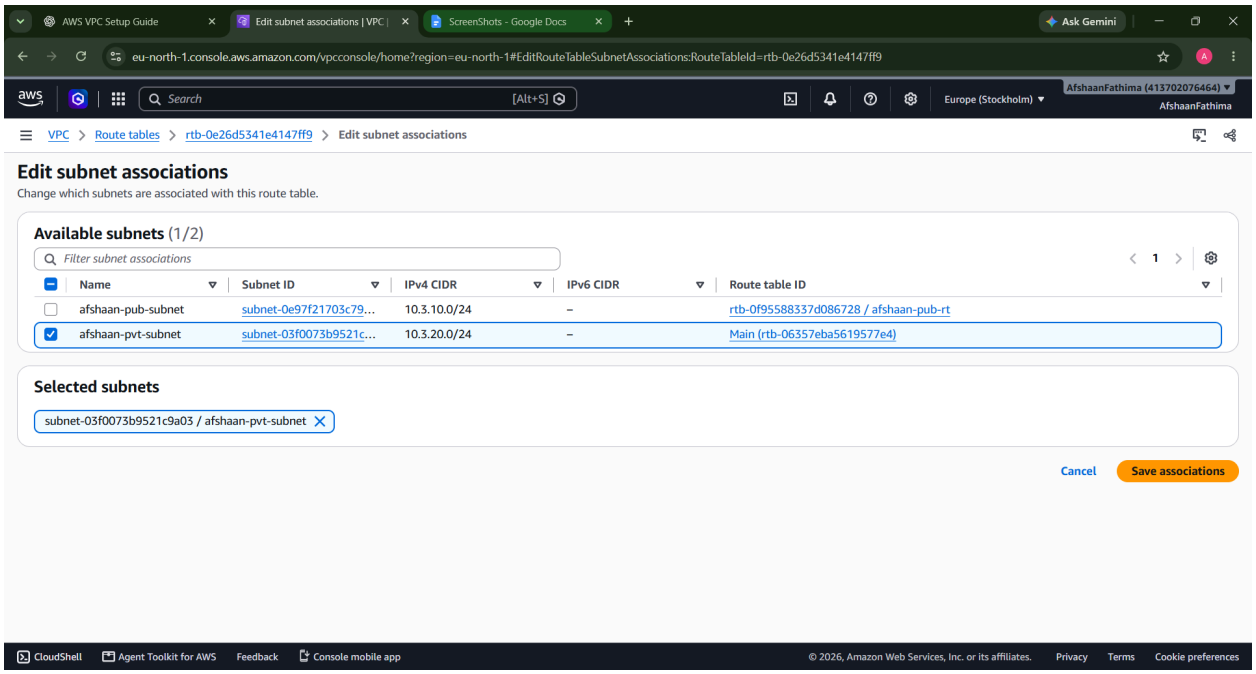
Create private route table – `afshaan-pvt-rt`



Add NAT Gateway route (`0.0.0.0/0`) to private route table

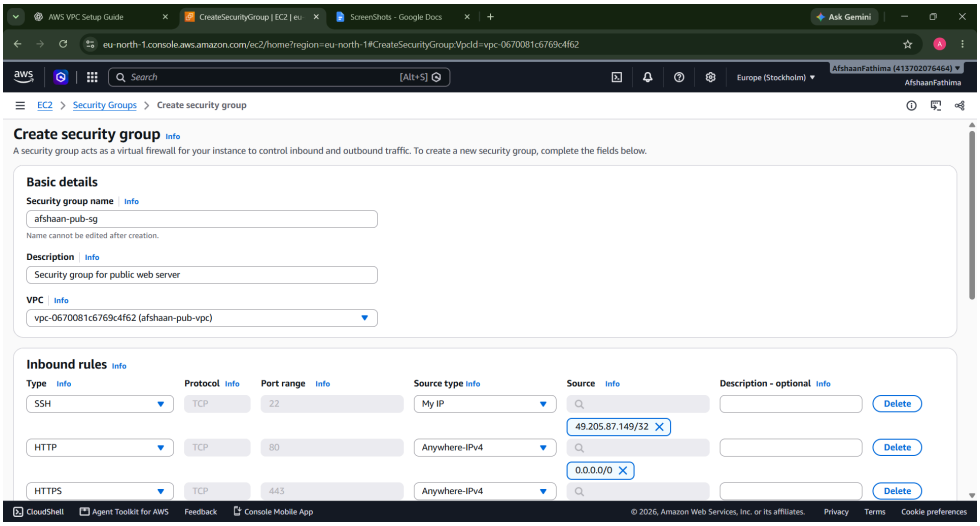


Associate private subnet with private route table



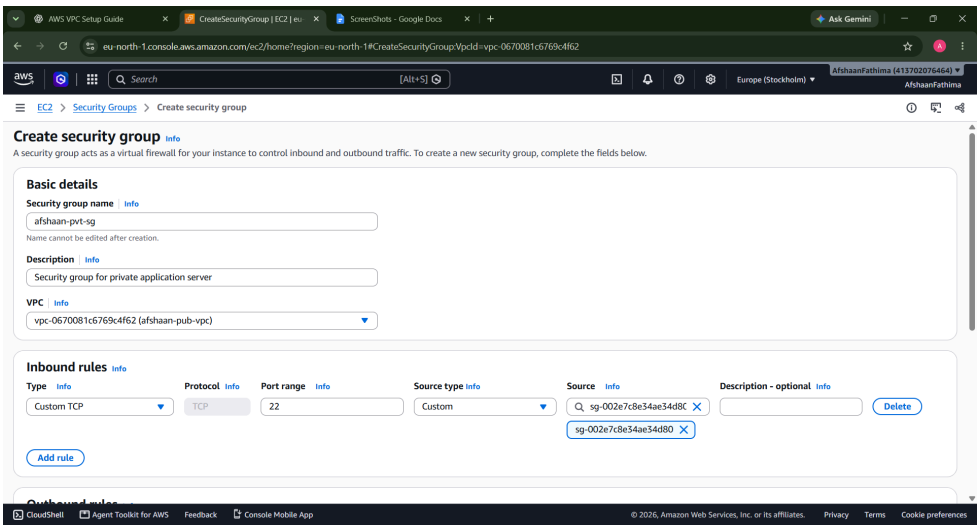
Security groups

Create public security group – `afshaan-pub-sg`

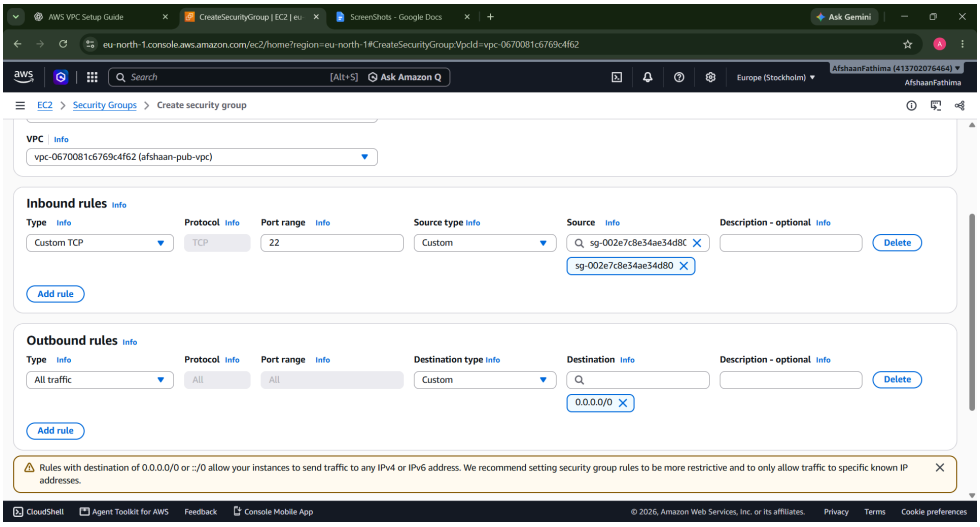


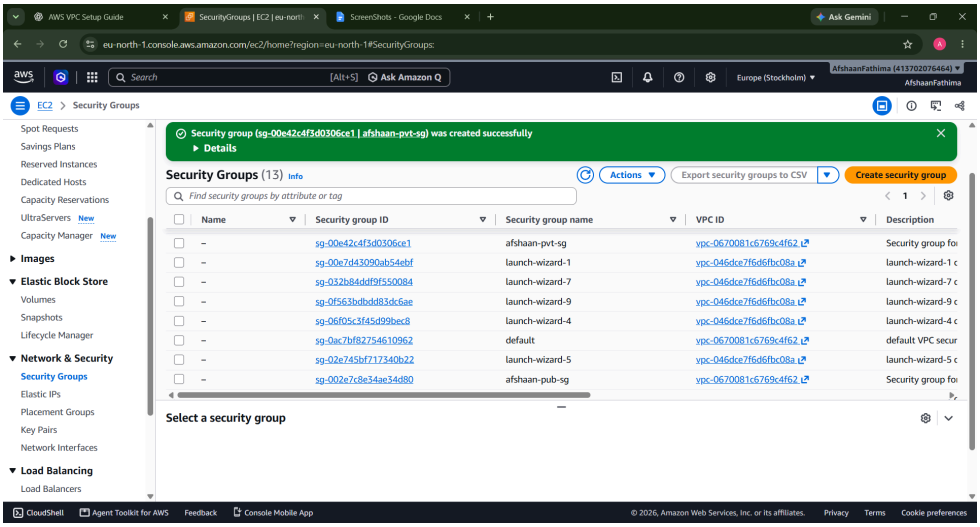
Allow SSH from your IP address

Create private security group – `afshaan-pvt-sg`



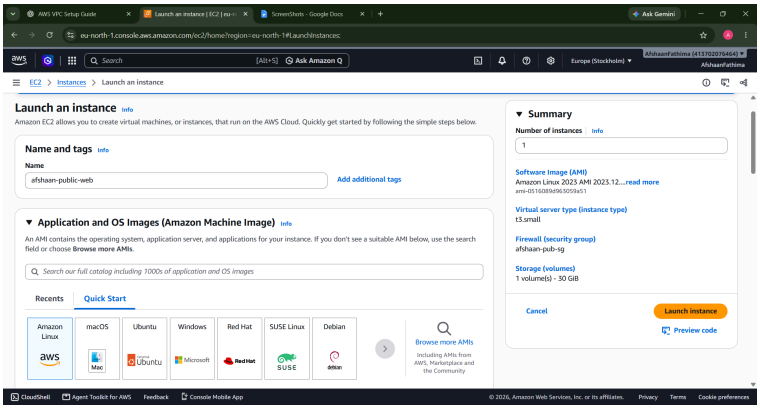
Allow application traffic only from the appropriate public-facing security group

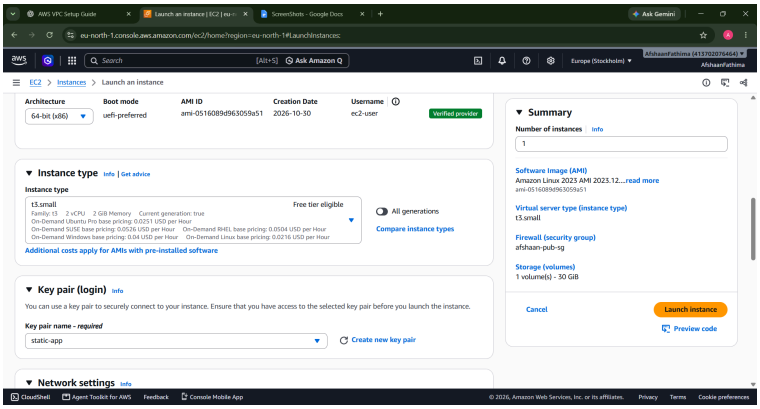


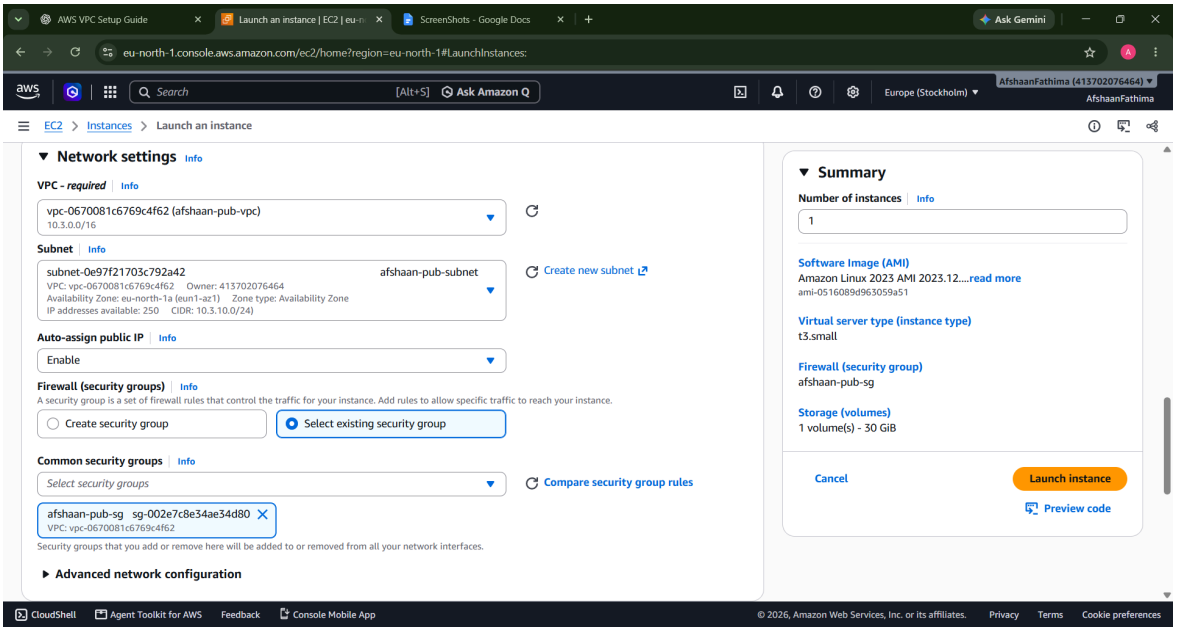


EC2 instances

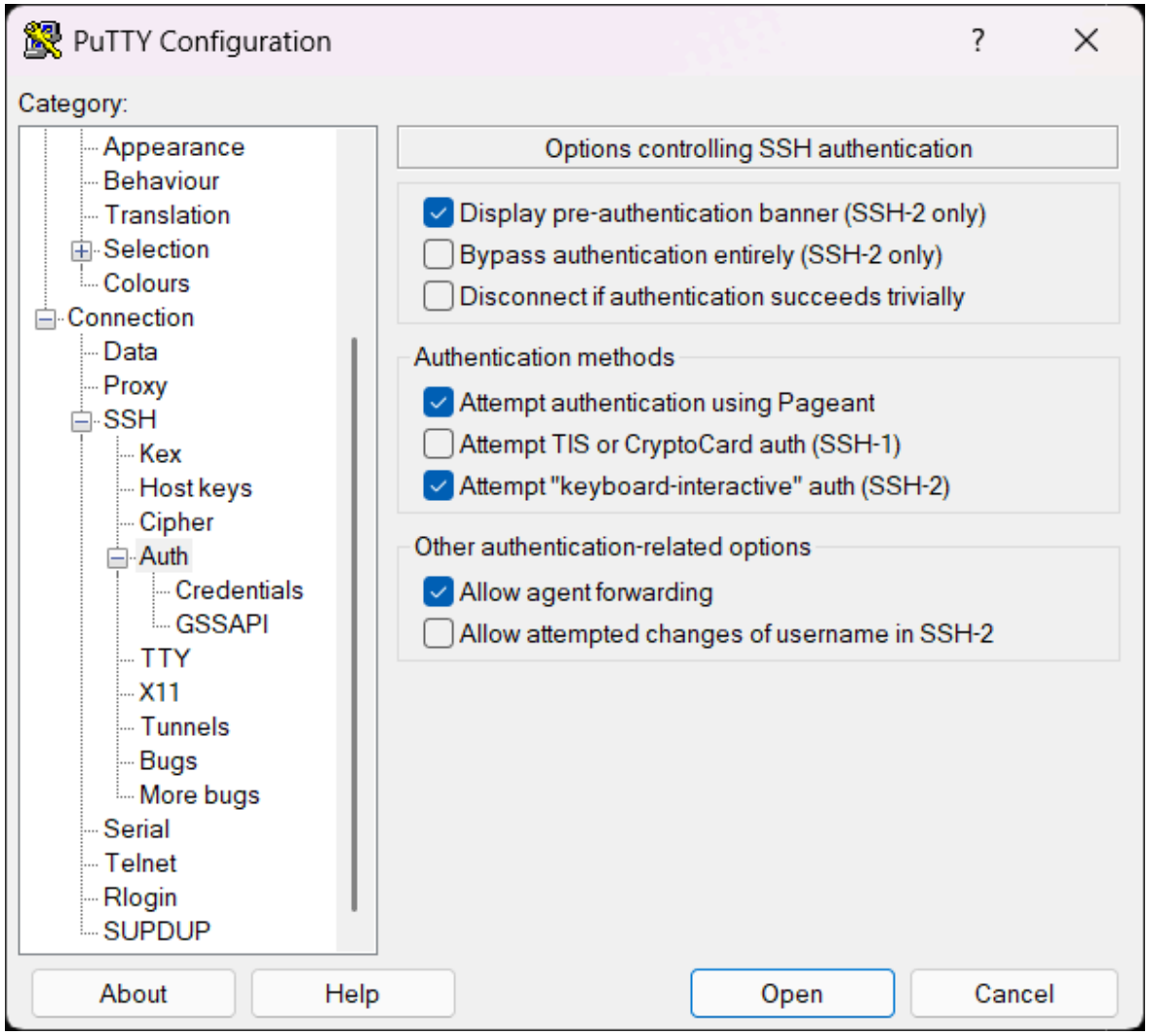
Launch public EC2 – `afshaan-public-web`

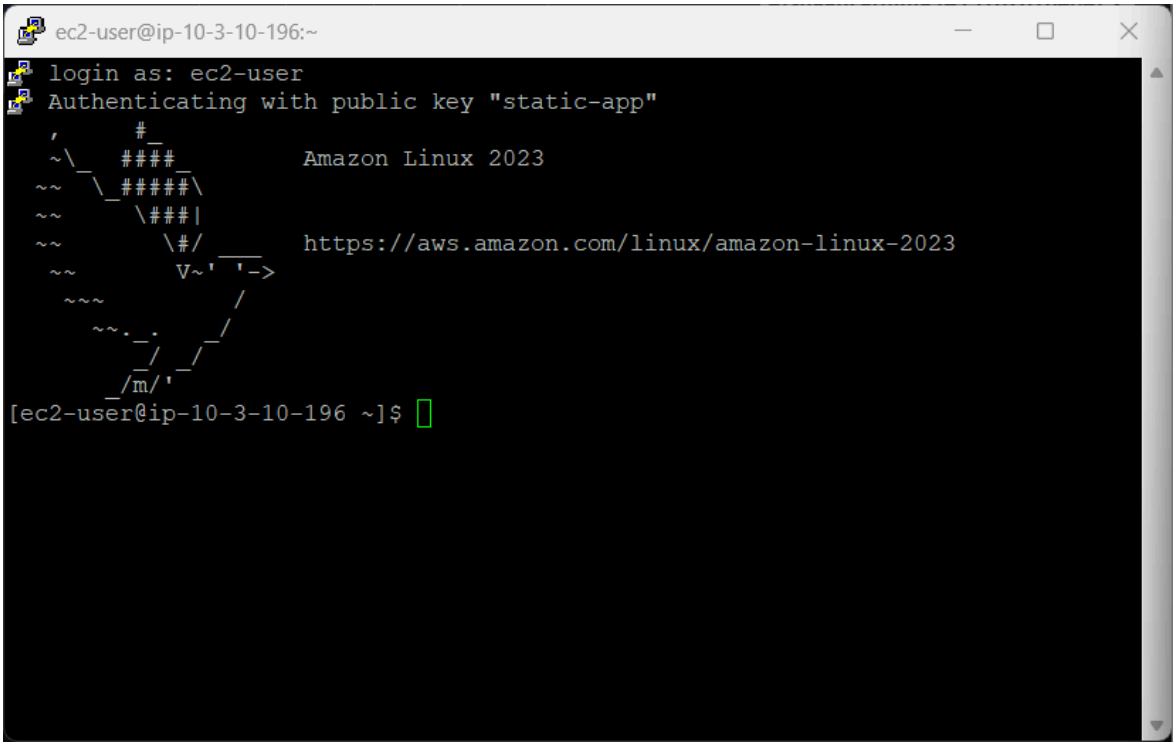




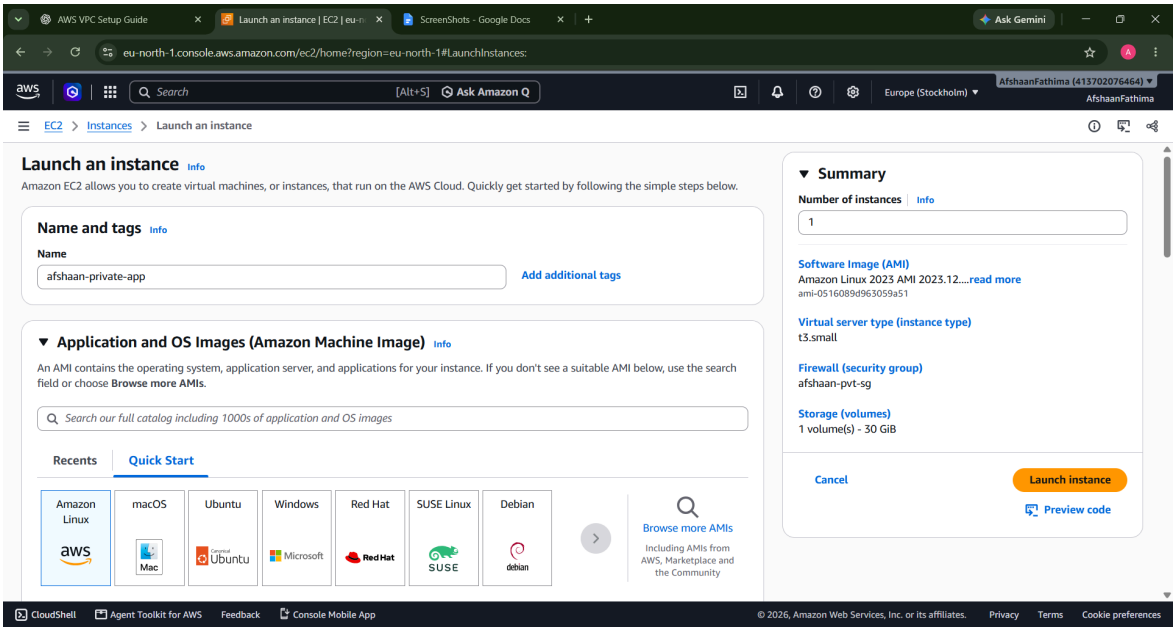


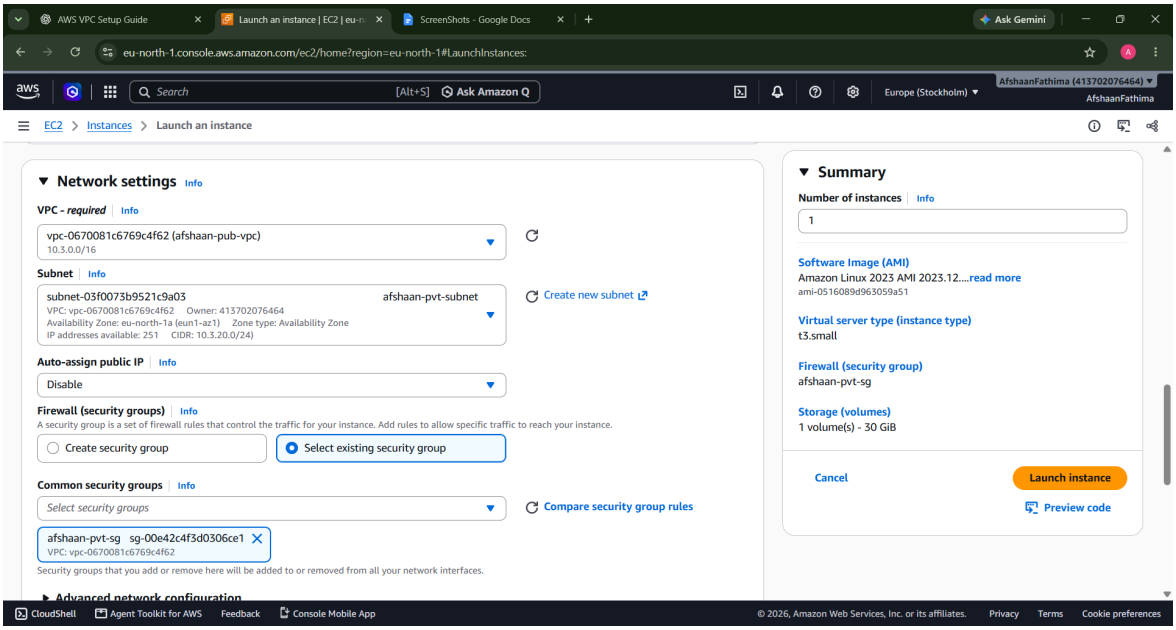
Connect to public EC2 using SSH


PuTTY Configuration
Category:
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Bugs
More bugs
Serial
Telnet
Rlogin
SUPDUP
Options controlling SSH authentication
Display pre-authentication banner (SSH-2 only)
Bypass authentication entirely (SSH-2 only)
Disconnect if authentication succeeds trivially
Authentication methods
Attempt authentication using Pageant
Attempt TIS or CryptoCard auth (SSH-1)
Attempt "keyboard-interactive" auth (SSH-2)
Other authentication-related options
Allow agent forwarding
Allow attempted changes of username in SSH-2
About
Help
Open
Cancel



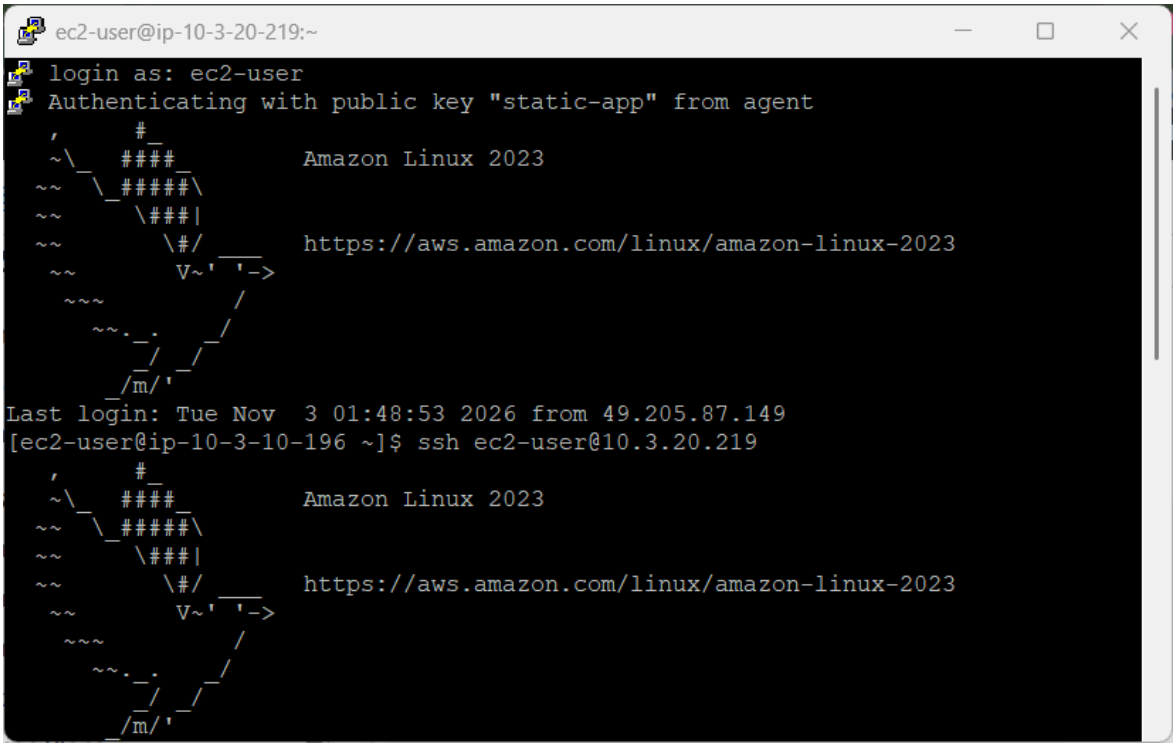
Launch private EC2 – `afshaan-private-app`





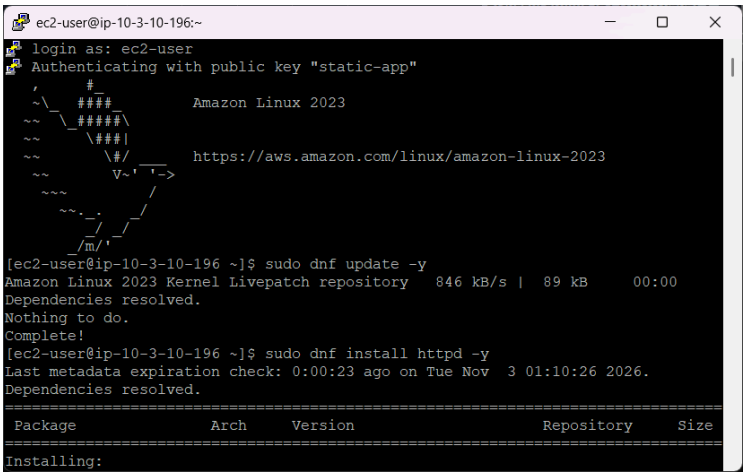
Disable public IPv4 assignment for private EC2

Connect to private EC2 through the public EC2 using SSH

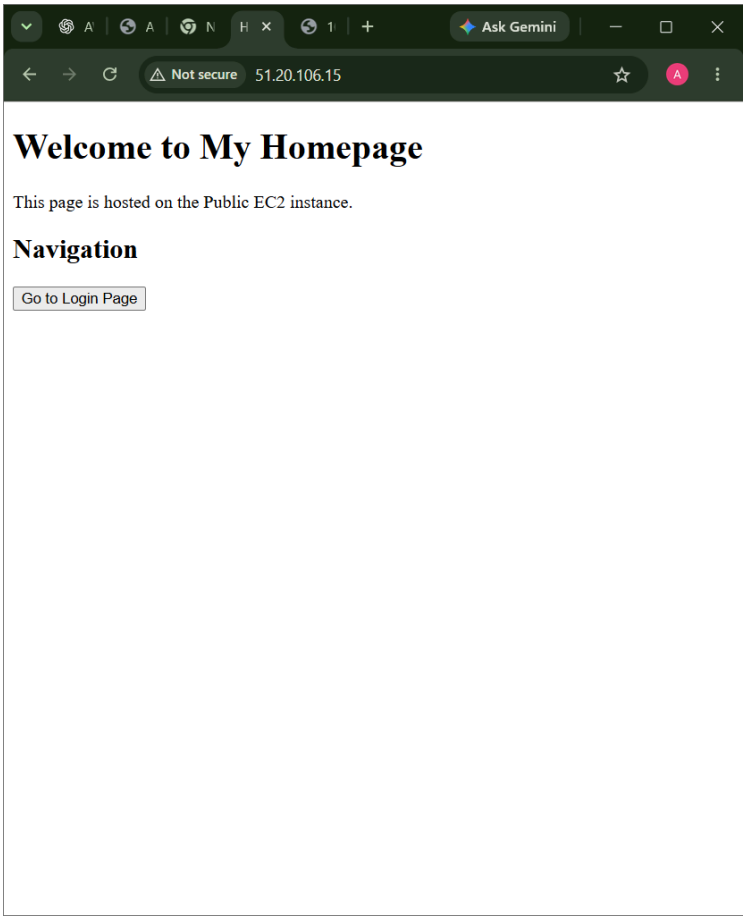


Phase 4 – Website and application

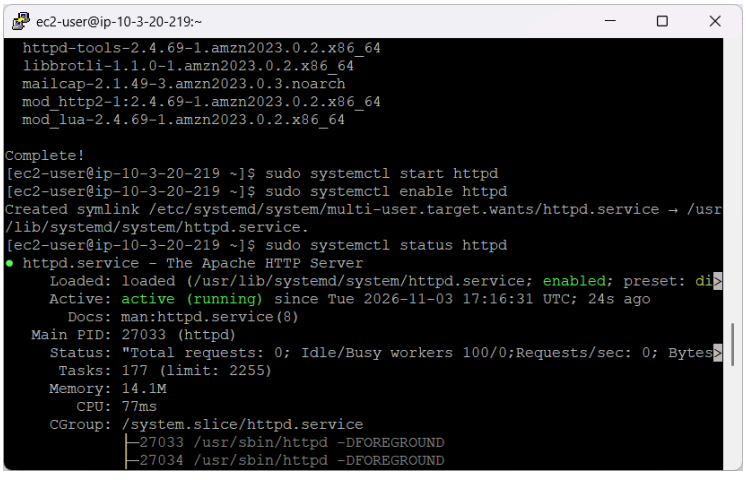
Install and configure a web server on the public EC2



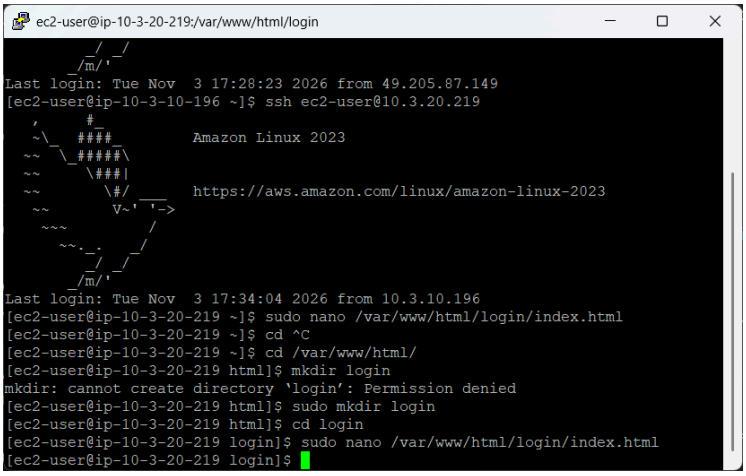
Create a simple public website



Install Apache on private EC2

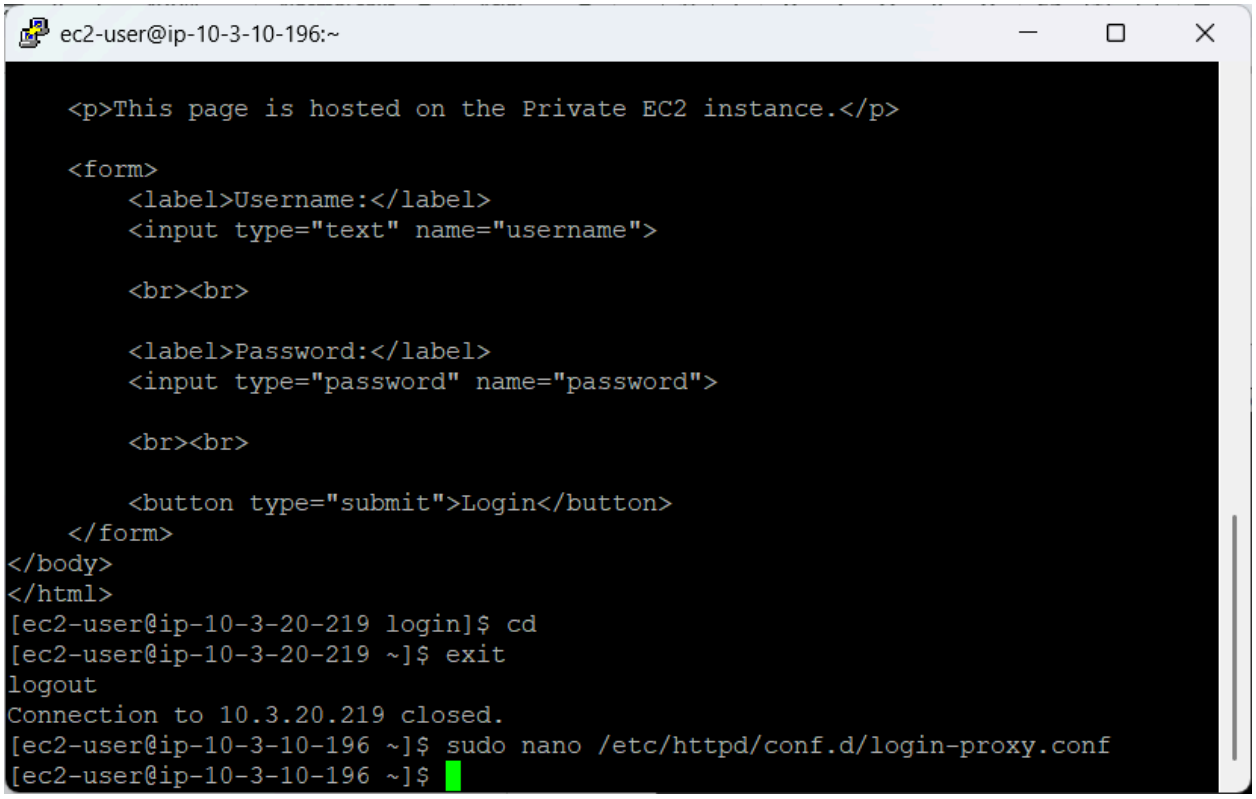


Create a simple login application on private EC2



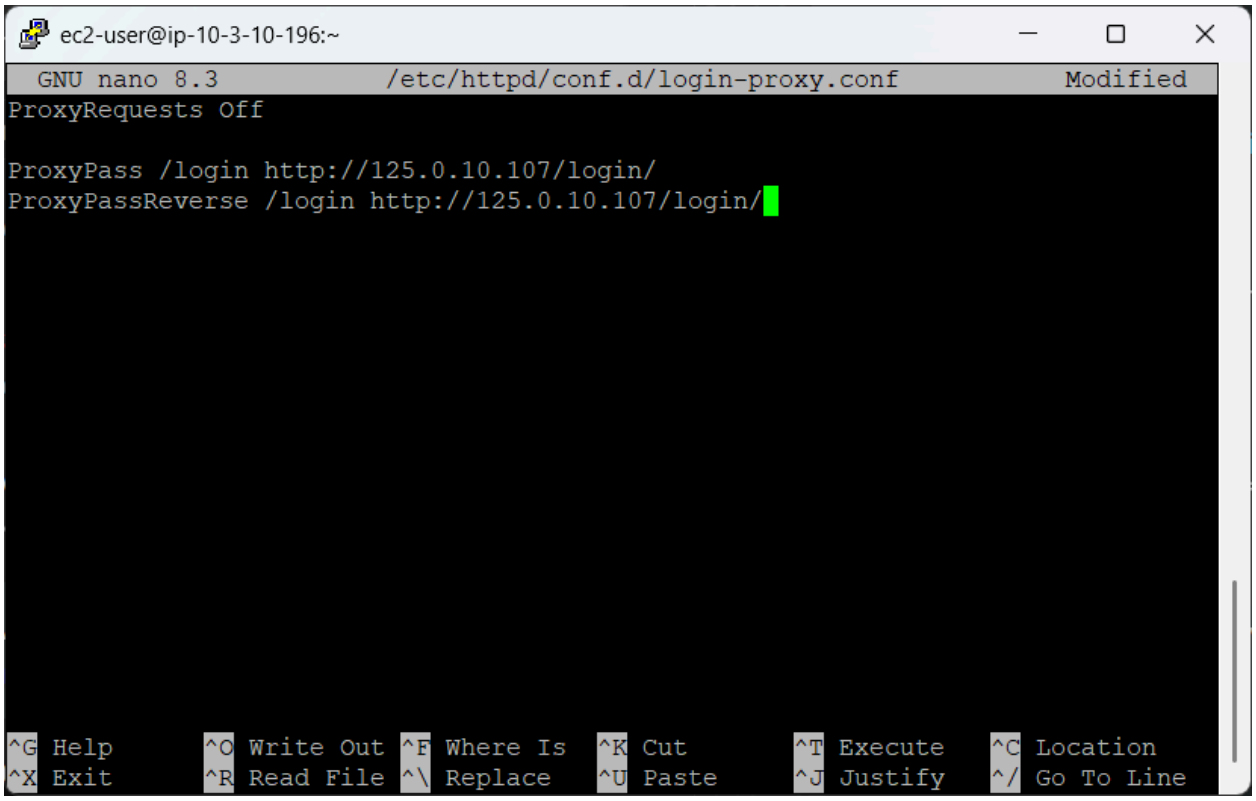
Test communication between public and private instances

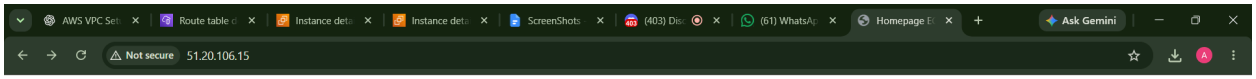
Phase 5- Apache proxy


ec2-user@ip-10-3-10-196:~
This page is hosted on the Private EC2 instance.
<form>
<label>Username:</label>
<input type="text" name="username">

<label>Password:</label>
<input type="password" name="password">

<button type="submit">Login</button>
</form>
</body>
</html>
[ec2-user@ip-10-3-20-219 login]$ cd
[ec2-user@ip-10-3-20-219 ~]$ exit
logout
Connection to 10.3.20.219 closed.
[ec2-user@ip-10-3-10-196 ~]$ sudo nano /etc/httpd/conf.d/login-proxy.conf
[ec2-user@ip-10-3-10-196 ~]$


ec2-user@ip-10-3-10-196:~
GNU nano 8.3 /etc/httpd/conf.d/login-proxy.conf Modified
ProxyRequests Off
ProxyPass /login http://125.0.10.107/login/
ProxyPassReverse /login http://125.0.10.107/login/
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

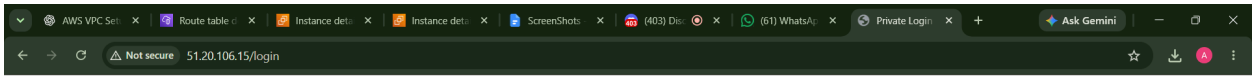


Welcome to My Homepage

This page is hosted on the Public EC2 instance.

Navigation

Go to Login Page



Welcome to the Private Login Server

This page is hosted on the Private EC2 instance.

Username:

Password:

Login